

Probabilistic Individual Short-Term Load Forecasting Using Conditional Variational Autoencoder

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Abstract—A novel probabilistic load forecasting method based on a Conditional Variational Autoencoder (CVAE) architecture is proposed in this paper. The proposed framework has been tested at the meter level, feeder level, and the Operating Center (“OC”, regional aggregate) level with real data. At the meter level, the proposed approach has been tested with data from 5000 different smart meters. Analysis at the feeder level included comparing the individual forecasts against the aggregate forecasts for 30 different feeders with a total over 16000 individual meters. At the OC level, the model was tested on 50 different OCs. According to the results, the proposed approach is capable of accurately and efficiently forecasting hourly load over a 4-day period at various levels of aggregation. To demonstrate the robustness of the proposed method, the proposed method has been tested on unseen customers from 10 different feeders with total number of 20880 customers. Results show that the proposed method is robust to unseen customers. Additionally, the proposed model has been compared with a range of benchmarks, including the well-established long short-term memory (LSTM) model and multiple linear regression (MLR). The CVAE algorithm is more accurate than the other algorithms in forecasting short-term loads.

Index Terms—Deep learning, individual short-term load forecasting, demand response, smart grid, smart meter.

Acronym	Explanation
CVAE	Conditional variational autoencoder
DL	Deep learning
ISTLF	Individual short-term load forecasting
LSTM	Long-short term memory
MAPE	Mean absolute percentage error
MLR	Multiple linear regression
OC	Operating center (regional aggregate)
RMSE	Root mean square error
sMAPE	Symmetric mean absolute percentage error

I. INTRODUCTION & LITERATURE SURVEY

Having an accurate short-term load forecast is essential for a variety of functions performed by electricity providers. ISTLF has at least two potential applications in smart grids. It allows for more detailed customer-specific operations, such as peak load shaving, and demand-side response, and can be used in demand response programs, particularly incentive-based demand response [1], [2]. The use of DL algorithms in short-term load forecasting has recently gained wide popularity. LSTM, one of the powerful DL methods, has been used for short-term load forecasting in recent years [1], [3]. However, LSTM based methods have a few drawbacks: 1) The process of training and readiness for real-world applications takes a lot of time and resources. 2) More importantly, LSTMs are prone to overfitting, making the dropout algorithm difficult to use

Parameters	Explanation
$\mathcal{X}$	Model Output
$\mathcal{Y}$	Model Input
D	Week indices
E	Historical energy for Lag time
H	Holiday indices
I	Interval indices
M	Month indices
T	Temperature
b	Batch size
$\theta_{\text{en}}, \theta_{\text{re}}$	Encoder Parameters
θ_{de}	Decoder Parameters
E_{train}	Number of epochs
N_s	Number of samples
N_t	Number of customers

[4]. Recently, researchers have also used more advanced DL methods, such as those proposed in [5]. They use temporal data pooling, to overcome overfitting issues, for training a variational recurrent deep embedding. An ISTLF forecasting technique based on deep residual networks was proposed in [6]. A Bayesian deep network approximation was proposed in [7], and [8] proposes an attentive transfer framework to address the ISTLF. A more comprehensive survey of short-term load forecasting is not attempted here, though it is recognized that this topic has a long history. This work focuses on improving the current understanding of the performance of CVAE-based architectures for ISTLF.

II. CONTRIBUTIONS

The limitations of existing methods in the literature and the datasets used in those works can be summarized as follows:

- The datasets used include a smaller customer count (typically a few hundred consumption time-series instances at most) or only have one type of load (i.e. commercial or residential)
- To the authors’ best knowledge, there are no studies that attempt the type of comprehensive performance analysis results provided in this paper.
- Most existing methods only provide a point forecast without providing an uncertainty level.

Therefore, motivated by these research gaps, a CVAE-based framework is proposed and extensively evaluated. Compared to the existing methods, the characteristics of the proposed method are

- It provides full distribution of the load at each timestamp. It is computationally efficient and easy to implement. Results in this paper represent 37% improvement over related literature in this area.
- Results are provided not just for larger (relative to earlier literature) datasets, but are evaluated using held-out customers that were not used in any way during model training or validation.
- In addition, this paper provides a comprehensive performance evaluation of the CVAE-based framework (in contrast to other VAE-based forecasting literature) by:
 - Evaluating the performance of the aggregates of the individual forecasts compared to forecasts at the aggregate level. Forecasts are aggregated up to feeder level as well to OC level.
 - Providing accuracy metrics across commercial and residential population strata.

The rest of this paper is organized as follows. Section III shows the problem statement and introduces the proposed framework. The performance evaluation is presented in Section IV, and the conclusions and future works are found in Section V.

III. PROBLEM STATEMENT AND SOLUTION FRAMEWORK

This section presents the features that are used in this paper for electrical load forecasting and pre-processing, followed by the evaluation metrics. The proposed probabilistic electrical load forecasting is then presented.

A. Feature Engineering and Data processing

Based on the dataset and testing different features, the input features used in this study are as follows:

- *Historical electrical load data*: sequence of energy consumption for the past Lag time steps, i.e. $\mathbf{E} = \{e_{t-\text{Lag}}, \dots, e_{t-2}, e_{t-1}\}$.
- *Temperature*: The temperature sequence is $\mathbf{T} = \{T_{t+1}, T_{t+2}, \dots, T_{t+N}\}$. In order to simulate uncertainties in forecast temperature values, 10% (of standard deviation) Gaussian noise is added.
- *Weekday*: Indices for different days are $\mathbf{D} = \{0, 1, \dots, 6\}$.
- *Month of the year*: Month indices are $\mathbf{M} = \{0, 1, \dots, 11\}$.
- *Interval*: Interval indices for one day with 24 intervals are $\mathbf{I} = \{0, 1, \dots, 23\}$.
- *Holidays*: Binary variable representing holidays, each of which can either be 0 or 1, i.e. $\mathbf{H} = \{0, 1\}$.

The raw data typically contains outliers and missing values due to problems such as the transmission interference of the devices, which can affect the efficiency of the forecasting model and result in a poor forecast. This necessitates the following processing of raw consumption data. To normalize the data the following equation was used

$$x_{i,\text{normal}} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, \quad (1)$$

where, x_i , $x_{\min}$, and $x_{\max}$ are the daily actual value, daily minimum, and daily maximum consumption. With normalized data, the input for the model is as follows:

$$\mathbf{Y} = \{\hat{\mathbf{E}}, \hat{\mathbf{T}}, \hat{\mathbf{I}}, \hat{\mathbf{D}}, \hat{\mathbf{M}}, \hat{\mathbf{H}}\}. \quad (2)$$

To train the model, the sequence of energy consumption's for next Lead time steps were used as labels;

$$\mathbf{X} = \{e_{t+1}, \dots, e_{t+2}, e_{t+\text{Lead}}\}. \quad (3)$$

Clustering users has been shown to improve the forecast accuracy, with comparing to have one global model over the whole meters, for example see [1]. In this study, the dynamic time warping time-series clustering method was used to cluster the dataset [9].

B. Proposed Method

Given both the input and output, the goal of CVAE is to fit a model of the conditional probability distribution $p_{\text{ap}}(\mathbf{X}|\mathbf{Y})$ [10], [11]. In this framework, an encoder with parameter θ_{en} gets the input $\mathbf{Y}$ and produces a Gaussian distribution $p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y})$. Before generating the posterior distributions, CVAE first projects the inputs into a lower dimensional space $\mathcal{G}$, called the latent space [12]. The latent space acts as a bottleneck to encourage the model to uncover the salient features [13]. The decoder, with parameters θ_{de} samples from latent space $\mathcal{G}$ and uses $\mathbf{Y}$ to generate the distribution $p_{\theta_{\text{de}}}(\mathbf{X}|\mathbf{Y}, \mathcal{G})$. Given these three components of the CVAE, the distribution of the output, $p(\mathbf{X}|\mathbf{Y})$, can be written as follows:

$$p(\mathbf{X}|\mathbf{Y}) \approx p_{\text{ap}}(\mathbf{X}|\mathbf{Y}) = \int_{\mathcal{G}} p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y}) p_{\theta_{\text{de}}}(\mathbf{X}|\mathbf{Y}, \mathcal{G}) d\mathcal{G}. \quad (4)$$

To produce $p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y})$, the samples from the $\mathcal{G}$ only should produce $\mathbf{X}$, i.e. $p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y}) \approx 0$. However, as shown in [12], this is an intractable problem due to the integral over the latent space variable $\mathcal{G}$ and a network cannot be trained directly to do this. One solution to this is to add a recognition network to the model which governed by trainable network parameters θ_{re} and create $\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})$. This function can help the model select values of latent space which are likely to produce $\mathbf{X}$. With the recognition network, the model has three sets of parameters. Overall, Θ_{cvae} (the CVAE model parameter), can be written as $\Theta_{\text{cvae}} = \theta_{\text{en}} \cup \theta_{\text{de}} \cup \theta_{\text{re}}$. The goal is to find the parameters of the network to maximize the log-likelihood of $p_{\text{ap}}(\mathbf{X}|\mathbf{Y})$ over the dataset [10]. Then (4) can be written as follows:

$$\begin{aligned} p_{\text{ap}}(\mathbf{X}|\mathbf{Y}) &= \int_{\mathcal{G}} \frac{p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y}) p_{\theta_{\text{de}}}(\mathbf{X}|\mathbf{Y}, \mathcal{G})}{\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})} \kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X}) d\mathcal{G} \\ &= \mathbb{E}_{\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})} \left(\frac{p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y}) p_{\theta_{\text{de}}}(\mathbf{X}|\mathbf{Y}, \mathcal{G})}{\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})} \right). \end{aligned}$$

To maximize the log-likelihood of distribution, it can be written as

$$\arg\max_{\Theta_{\text{cvae}}} \log p_{\text{ap}}(\mathbf{X}|\mathbf{Y}) =$$

$$\arg\max_{\Theta_{\text{cvae}}} \log \mathbb{E}_{\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})} \left(\frac{p_{\theta_{\text{en}}}(\mathcal{G}|\mathbf{Y}) p_{\theta_{\text{de}}}(\mathbf{X}|\mathbf{Y}, \mathcal{G})}{\kappa_{\theta_{\text{re}}}(\mathcal{G}|\mathbf{Y}, \mathbf{X})} \right).$$

A variational lower bound of the log-likelihood can be used as a surrogate objective function in the stochastic gradient variational Bayes framework to estimate the CVAE parameters [14]. By using Jensen's inequality, the evidence lower bound (ELBO) is given as

$$\begin{aligned} \log p_{\text{ap}}(\mathbf{X}|\mathbf{Y}) &= \\ &- \mathbb{E}_{\kappa_{\theta_{\text{re}}}(\cdot|\cdot, \cdot)} \left[p_{\theta_{\text{de}}}(\cdot|\cdot, \cdot) \right] + \mathfrak{C} \left(\kappa_{\theta_{\text{re}}}(\cdot|\cdot, \cdot) \parallel p_{\theta_{\text{en}}}(\cdot|\cdot) \right) \geq \text{ELBO}, \end{aligned}$$

where $\mathfrak{C}(\cdot \parallel \cdot)$ is the Kullback-Leibler divergence between two distributions and can not be negative. Now, the ELBO can be optimized via stochastic gradient descent [10]. The proposed algorithm for training the CVAE model is presented in Algorithm 1 and testing procedure is provided in Algorithm 2. Fig. 1 illustrates the graphical model of a CVAE during the training and test phases. The parts of the training which are not in the testing phase, are shown in dim color.

Algorithm 1: Training the proposed model.

```

1 Clean the data, and get the features from the data, and
  cluster the data based on Section III-A, normalize the data
  based on (1), build the input and labels for training the
  model based on (2) and (3)
2 while epoch  $\leq E_{\text{train}}$  do
3   while  $k \leq I_t$  do
4     while  $b \leq B_t$  do
5       Feed the main encoder with batch  $b$  of  $\mathcal{Y}$  Feed
        the reconstruction encoder with batch  $b$  of  $\mathcal{Y}$ ,
        and  $\mathcal{X}$ 
6       Calculate the Kullback-Leibler distance between
        two distributions  $\mathcal{N}(\mu_{\text{re}}, \Sigma_{\text{re}})$  and
         $\mathcal{N}(\mu_{\text{en}}, \Sigma_{\text{en}})$ 
7       Feed the decoder with a batch  $b$  from  $\mathcal{Y}$ 
8        $b \leftarrow b + 1$ 
9     end while
10    Update the  $\Theta_{\text{cvae}}$ 
11     $k \leftarrow k + 1$ 
12  end while
13  epoch  $\leftarrow$  epoch + 1
14 end while

```

Algorithm 2: Testing the trained model.

```

1 Clean the data, and get the features from the data based on
  Section III-A, normalize the data based on (1), build the
  input for testing the model based on (2) as  $\mathcal{Y}_t$ 
2 for  $1 \leq i \leq N_t$  do
3   Feed the main encoder with batch  $b$  of  $\mathcal{Y}_t$ 
4   Draw a latent space sample at random from the latent
    space distribution defined by  $\mathcal{N}(\mu_{\text{en}}, \Sigma_{\text{en}})$ 
5   Feed the pre-trained decoder network with the drawn
    sample and corresponding original  $\mathcal{Y}_t$  data
6   Draw a random sample with  $N_s$  from the  $p(\mathcal{X}_p | \mathcal{Y}_t)$  as
    the predicted values.
7 end for

```

C. Evaluation Metrics

This study considers three metrics to evaluate the model namely, RMSE, MAPE, sMAPE [1].

IV. CASE STUDIES

In this section, numerical results for the different levels (meter, feeder, OC) are provided in more detail. In this paper, convolution layer was used as the input layer, with kernel size = 10, stride = 10. The other hyperparameters of the model are latent space size that is set as 100, the learning

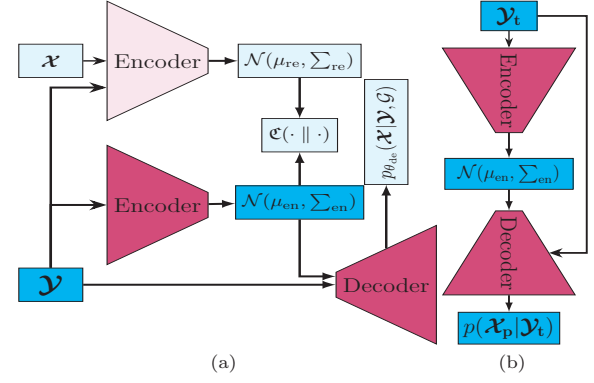


Fig. 1. The architecture of the proposed CVAE: (a) during training; (b) during test.

rate is set to 2×10^{-5} . ReLU used as activation function for the model. The tslearn.clustering package [15] was used for clustering the data and sequence of energy consumption (Lag) for the input set to seven days. The sequence of the energy consumption for prediction at each window (Lead) is set to one day. All experiments were conducted in Python 3.8 using the PyTorch [16] framework. A well provisioned GPU instance on the Oracle Cloud Infrastructure was used to train and test the model.

To generate the dataset, 5000 meters were randomly selected from an electric utility serving the midwest US. Data were collected for one year (January 2019 to January 2020). For the feeder level, 30 feeders were randomly selected (with the total number of 16564 meters), with one year of historical data. Time intervals were restricted from 12 PM to 7 PM, which covers the hours of the day when demand response events are likely to occur. In all test cases, the load is predicted for the next four days and in the peak load prediction cases, the peak load of each day over the next four days was considered. Using the Elbow method, the optimal number of clusters was determined. Based on the distortion versus the number of clusters, the optimal number of clusters was set to 15 for ISTLF and 6 for OCs (the number of clusters differed for different experiments).

A. Performance of Individual Meter Forecast

In this section, the results for the 5000 individual meters are provided. Fig. 2 shows the distributions for three evaluation metrics over 5000 meters (against held-out test data). The mean of the metrics are shown in Table I. Fig. 3 shows the probabilistic forecasts and the actual value with 95 % confidence intervals for one specific meter for the next four days. The mismatches in Fig. 3 between the actual values and the predicted values verify that the model is well-trained and has acceptable accuracy.

TABLE I
MEAN OF METRICS FOR 5000 METERS.

Metric	RMSE	sMAPE (%)	MAPE (%)
Value	0.11	29.32	27.57

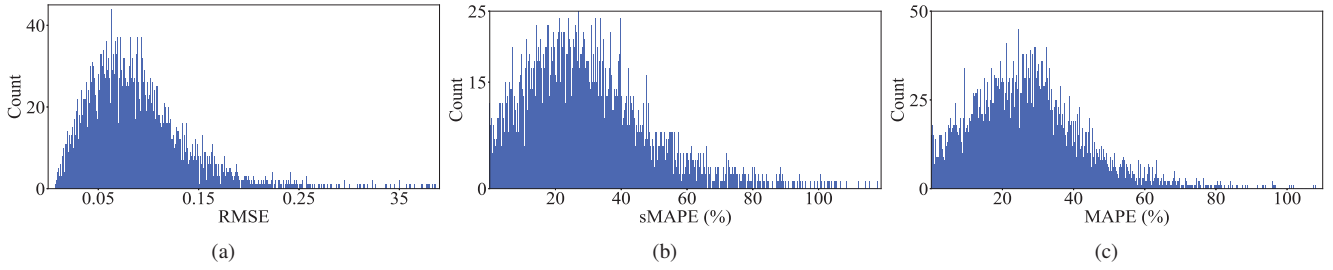


Fig. 2. Distribution of the evaluation metrics for 5000 meters; (a) RMSE, (b) sMAPE, (c) MAPE.

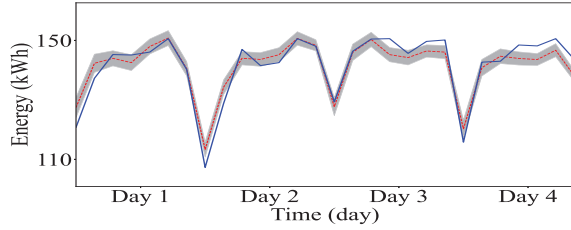


Fig. 3. Probabilistic forecasts and the actual value of one specific meters with 95 % confidence intervals; - Real value, - - Predicted value.

B. Commercial vs Residential

Compared to residential loads, commercial loads have more stable load profiles, and their daily on/off patterns make them more predictable in some cases. There are of course, various examples of variation from this observation; load can vary based on business outlook, demand/supply etc. These regressors are beyond the scope of this study. Since some of the commercial loads depend on the season or the month, prediction for each day's peak consumption was also evaluated. Table II shows the average of metrics for all commercial loads versus residential loads. As can be seen (from MAPE and sMAPE metrics: RMSE is based on kWh values, and since commercial loads have higher values than residential loads, the RMSE will be higher, even if the model is more efficient.), the trained model performs better on commercial loads when the model predicts over the whole period, however, for peak load prediction, the model performs better on residential loads.

TABLE II
METRICS FOR COMMERCIAL AND RESIDENTIAL LOADS.

Metric	Whole period		Peak	
	Resid.	Comm.	Resid.	Comm.
RMSE	0.05	0.95	0.04	0.10
sMAPE (%)	29.72	23.53	23.12	25.30
MAPE(%)	27.75	25.09	27.96	30.27

C. Comparison with Existing Literature

The best reported MAPE in [1] is 44.06 %. It is important to note that the MAPE for all customers in the proposed approach is 27.57 %, which is 37 % smaller than the best-reported MAPE in [1]. While this may be due to differences in choice and quality of data, these results are based on actual

anonymized data, and a considerably larger dataset, so can be considered more representative of typical load and model behavior.

D. Forecasting Accuracy for Unseen Customers

In this section, the results from testing the proposed method on a sample of unseen customers are provided. Ten different feeders are selected for this purpose. Initially, the model was trained with one feeder's customers, and then tested with the remaining nine feeder's customers. For example, the first model was trained with customers of feeder 1 and then tested with customers of {feeder 2, feeder 3, ..., feeder 10}, and the second model was trained with customers of feeder 2 and was tested with customers of {feeder 1, feeder 3, ..., feeder 10} and so on. A total of 10 models were tested with customers from 90 feeders (with a total of 20880 customers). The other settings (such as the number of prediction days, etc.) are the same as in the other sections. Table III shows the mean of MAPE for individual forecasts, MAPE of the aggregate of forecasts, and forecast of the aggregate for all customers. It is evident that the proposed model can effectively forecast the load at the individual meter level, as well as at the aggregate level for unseen customers.

TABLE III
MEAN OF MAPE FOR UNSEEN CUSTOMERS (TOTAL COUNT: 20880).

Metric	Individual forecasts	Forecast the aggregate	Aggregate the forecasts
MAPE(%)	36.97	9.21	7.20

E. Comparison with other Benchmark Methods

In this section, different benchmark methods are implemented on the dataset and the results are compared. The MLR [17], and LSTM [1] are implemented on the dataset with the same features. Table IV shows the mean of metrics for all meters for these three methods. As it can be seen, the proposed method has better performance than the other two methods.

F. Performance of Feeder Level Forecast

In order to forecast at the feeder level, it is possible to either forecast first, then aggregate, or aggregate first, then forecast. To compare aggregating the forecasts versus forecasting the aggregate, the model was trained for each meter connected

TABLE IV
MEAN OF METRICS FOR LSTM, MLR, AND PROPOSED METHOD.

Method	RMSE	sMAPE (%)	MAPE (%)
MLR	0.16	38.25	41.15
LSTM	0.13	36.60	34.70
Proposed	0.11	29.32	27.57

to each feeder, and also trained separately with the aggregate consumption of the meters. Table V shows the average of metrics for all 30 feeders with total number of 16564 meters for each scenario. As can be seen, based on the mean of evaluation metrics for all feeders, aggregating the forecasts provides more accurate forecasts than forecasting the aggregate for whole day prediction and also peak prediction. Individual forecast errors canceling out each other may partially explain the better performance achieved by aggregating the forecast.

TABLE V
MEAN OF METRICS FOR WHOLE DAY FOR ALL 30 FEEDERS.

Prediction	Aggregate the forecasts			Forecast the aggregate		
Metrics	RMSE	sMAPE(%)	MAPE(%)	RMSE	sMAPE(%)	MAPE(%)
Whole	0.07	6.84	6.54	0.09	8.54	8.53
Peak	0.10	6.60	6.30	0.11	9.26	8.74

G. Performance for OC Level

In this study, 50 OCs with total number of 1733407 customers were considered. However, the aggregated load of each OC was used to train and test the model. Two different approaches were applied for prediction; 1) The OCs were clustered into six classes based on their load pattern, and each class was trained separately, 2) each OC was trained separately. Table VI shows the mean of three metrics for the all OCs level data with two approaches. As discussed before, consumption behavior of individuals is subject to greater uncertainty because of the influence of various internal and external factors. Thus, load forecasting is much easier at the aggregated level, and error of prediction decreases as more consumers are aggregated in general. As shown in [18], for different forecasting methods, aggregating more customers improves relative forecasting performance. As forecasting levels change or aggregation goes up the system hierarchy e.g. individual to the feeder, feeder to OC, this improvement becomes more apparent (It can be seen this from the Tables I, V, and VI). However, it can not be concluded that feeders (at the same aggregation level) with more customers will necessarily have lower prediction errors when compared to feeders with fewer customers.

TABLE VI
MEAN OF METRICS FOR ALL OCs.

Metrics	RMSE	sMAPE (%)	MAPE (%)
With clustering	0.05	4.95	4.98
Without clustering	0.04	2.61	2.62

V. CONCLUSIONS AND FUTURE WORK

This paper proposes a Deep Learning-based model (CVAE) for forecasting short-term electrical loads for residential households and commercial accounts trained on real-world consumption data. Insightful case studies at different levels are presented and discussed in details which show the effectiveness and robustness of the proposed method. This work in this space has thrown open a number of questions that deserve additional research; 1) Finding the right features or architectures to improve the accuracy for commercial peak prediction. 2) Exploring how the delta between forecast the aggregates and aggregating the forecasts behaves with more historical data.

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